Oluwatosin Ayeni

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G-Research, London, UK

Dear Hiring Team,

I am writing to apply for the NLP Performance Engineer role at G-Research. With hands-on production experience in vLLM, SGLang, TensorRT-LLM, and CUDA kernel profiling, I am directly aligned with the technical depth this role demands.

Three achievements most relevant to your requirements:

- **H100 NVL benchmark — vLLM vs SGLang:** I benchmarked vLLM v0.30 against SGLang v0.5.20 on 2× NVIDIA H100 NVL 94 GB (NVLink 4.0, CUDA 13.4) running Qwen3-8B in BF16 with tensor parallelism 2. Peak throughput: 5,965 tok/s; 100% SLO compliance from concurrency 1 to 128; 1,443× TTFT speedup via RadixAttention prefix caching (39 ms vs 55,565 ms at 131K-token context). Live at inference-web-theta.vercel.app/benchmarks/llm.
- **CUDA-level KV cache optimisation:** Using NVIDIA Nsight Systems and Compute, I identified a memory bandwidth bottleneck in KV cache ops on RTX 4090 hardware and tuned the kernel access pattern, delivering a 3.8× throughput improvement without hardware change. I followed this with an 18× inference latency reduction through continuous batching redesign.
- **Production inference gateway at scale:** I built *inference-optimizer* — implementing parallel execution (3.87× faster), prompt caching (61.4% cost reduction), model routing (64.3% cheaper), and concurrency management (14× throughput) across Claude, GPT-4o, and Gemini.

G-Research's reputation for hiring engineers who combine rigorous quantitative thinking with systems depth is exactly the environment I want to work in. The intersection of NLP at scale and GPU performance engineering is where I have focused my independent work for the past two years.

I am happy to walk through the benchmarks in detail, provide compiled materials, or discuss my approach to GPU profiling and serving stack design. I look forward to speaking.

Yours sincerely,

Oluwatosin Ayeni

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